

Multiple Choice

1. **Answer:** D

Options: A) Melting point; B) Boiling point; C) Polarizability; D) First ionization energy

Note: Correct option: D - First ionization energy

2. **Answer:** B

Options: A) The I⁻ : IO₃⁻ ratio is 3:1.; B) The MnO₄⁻ : I⁻ ratio is 6:5.; C) The MnO₄⁻ : Mn₂⁺ ratio is 3:1.; D) The H⁺ : I⁻ ratio is 2:1.

Note: Correct option: B - The MnO₄⁻ : I⁻ ratio is 6:5.

3. **Answer:** B

Options: A) 23.56 GHz; B) 42.58 MHz; C) 74.34 kHz; D) 13.93 MHz

Note: Correct option: B - 42.58 MHz

4. **Answer:** A

Options: A) HCl; B) AgCl; C) CaCl₂; D) CCl₄

Note: Correct option: A - HCl

5. **Answer:** D

Options: A) BeCl₂; B) MgCl₂; C) ZnCl₂; D) SCl₂

Note: Correct option: D - SCl₂

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Most organic compounds do not contain silicon atoms.'

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Each hybrid orbital may hold four electrons.'

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.050 M AlCl₃'.

Long Form Response

11. **Answer:** Ca. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** H₃O⁺ / H₂O. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key